

PHOSPHORIC ACID 81%

Description

Phosphoric Acid 81% is a clear, food-grade solution produced from high-purity phosphoric acid. It is commonly used in brewing for yeast acid washing and for lowering alkalinity in brewing liquor.

Benefits

Removes bacterial contaminants from yeast slurries.
Lowers alkalinity in brewing liquor without altering the chloride/sulphate balance.

Application and Dosage Rates

REFER TO THE SAFETY DATA SHEET BEFORE USING THIS PRODUCT.

BREWERY USAGE

Phosphoric acid may be added to either the cold or hot liquor tank and must be mixed thoroughly. Allow time for carbon dioxide to escape as excess carbonate is neutralised. When refilling your tank, consider any remaining treated liquor as this will influence alkalinity levels.

Dosage requirements depend on the alkalinity of the untreated liquor and the desired water profile or beer style.

Adding 5 mL of Phosphoric Acid 75% per hL of water reduces alkalinity by 30.2 mg/L (ppm).

Note: Phosphoric acid can react with calcium, forming precipitates that may reduce calcium levels in the mash.

Phosphoric Acid for Acid Washing

Acid washing is used by brewers to reduce bacterial contamination in pitching yeast. When performed correctly, it eliminates bacteria with minimal impact on yeast health. The effectiveness of the process depends on maintaining the correct temperature, pH, and contact time.

NOTE: Acid washing does not remove wild yeast.

Instructions for Acid Washing Yeast

- Cool both the yeast and the diluted (1:10) food-grade phosphoric acid 81% to 2–5°C and maintain this temperature throughout the process
- Measure the required amount of yeast for brewing and transfer it into a sanitised container
- Begin acid washing 60–90 minutes before pitching by thoroughly mixing the diluted phosphoric acid into the yeast slurry until a pH of 2.0–2.3 is reached
- Hold the yeast at this pH for a maximum of 60–90 minutes, stirring continuously

PRODUCT
CODE

7507

PACKAGING

25 Litres

STORAGE

Store in original
container.

Keep away from
Chlorines.

Recommended
storage
temperature 10 -
15 Degrees C.

Store away from
direct sunlight.

Do not allow to
freeze.

SHELF LIFE

Use within 12
months from the
date of
manufacture.

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- Pitch the entire mixture into the fermenter with wort

Do not wash yeast on a non-brewing day. Viability decreases significantly with storage time.

DON'TS of Acid Washing:

- Don't deviate from the recommended temperature, time, or pH limits
- Don't store washed yeast
- Don't wash compromised yeast (e.g., long-stored yeast, heavily contaminated yeast, or yeast from slow fermentations)
- Don't wash yeast from very high-gravity fermentations (>8% v/v ethanol)

Common Faults When Acid Washing and Their Solutions

- Yeast slurry temperature too high: Reduce contact time between acid and yeast
- Yeast slurry pH too low: Reduce contact time or add additional yeast
- Yeast slurry temperature too high: Reduce contact time between acid and yeast

Guideline for Use

- Confirm suitability for the intended application
- In case of skin or eye contact, rinse immediately with plenty of water
- Wash away any spillages with ample water
- Read the Safety Data Sheet before use

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